2627-MA140: Week 01, Lecture 3 (L03)

Polynomials and Rational Functions

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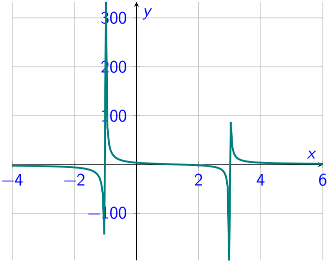


Today we rationalize the following topics:

1. News!
   - Tutorials
   - Tutorial sheet
2. Polynomials
   - Quadratic
   - Sketching polynomials
3. Rational Functions
   - Long division
4. Towards Partial Fractions
5. Exercises

See also Sections 1.2 and 7.4(!) of `https://math.libretexts.org/Bookshelves/Calculus/Calculus_(OpenStax)/01%3A_Functions_and_Graphs`

Slides are on canvas, and at `https://www.niallmadden.ie/2627-MA140/`

News!

Tutorials start **next** week. The schedule is:

Teams	Time	Venue	Leader
1, 2	Tuesday 15:00	ENG-**2003**	ST
3, 4	Tuesday 15:00	ENG-**2034**	DC
9, 10	Thursday 11:00	ENG-**2002**	ST
11, 12	Thursday 11:00	ENG-**3035**	DC
5, 6	Friday 13:00	Eng-**2002**	ST
7, 8	Friday 13:00	Eng-**2035**	DC

Do these data agree with information you have been provided?

Would you be interested to taking a tutorial through Irish? (Show of hands?)

You don't have to complete a graded assignment next week. However, this is a "practice" one available. See https://universityofgalway.instructure.com/courses/58477/assignments/162051

During tutorials, the tutor will solve some similar questions. You can access the **tutorial sheet** at the link above (and on Canvas: `Modules`... `Tutorial Sheets`).

The Tutorial Sheet has questions that are nearly identical to your own version.

Polynomials

Yesterday, we saw that...

Polynomials

Polynomials are functions of the form

$$y = f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x^1 + a_0, \quad x \in \mathbb{R},$$

where $a_0, a_1, ..., a_n$ are real numbers called the **coefficients** of the polynomial. The number n is called the **degree** of the polynomial.

Examples:

Example: quadratic

$x^2 - 2x - 3$ is a **quadratic** polynomial: it has degree $n = 2$.

There are many occasions when we want to **factorise** such quadratics, meaning we write them as the product of a pair of linear polynomials.

For example, we can **factorise** $x^2 - 2x - 3$ as

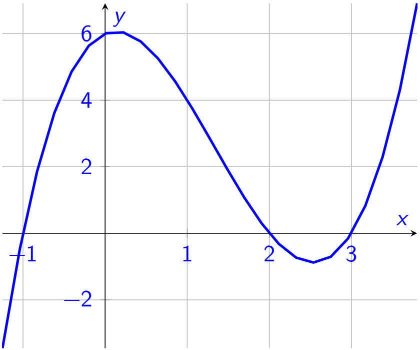
$$x^2 - 2x + 3 = (x - 3)(x + 1)$$

It is important to note that not all quadratic polynomials can be factorised as two linear polynomials. Such quadratics are called **irreducible**.

For example, $x^2 + 1$ is irreducible.

Example

$y = x^3 - 4x^2 + x + 6$ is a **cubic** function with degree $n = 3$.



Fact

A polynomial function of degree n has **up to** $n-1$ turning points ("bends").

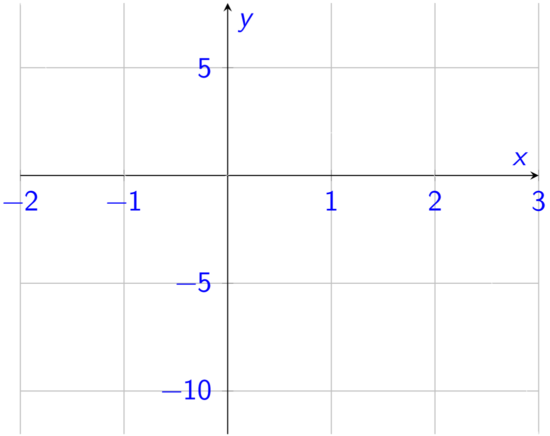
Examples:

When sketching the graph of a function, we first find the **intercepts**:

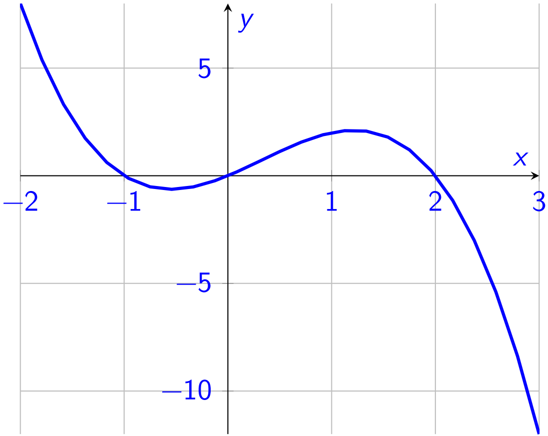
- The y-**intercept** is where the graph of the function cuts the y-axis: found by letting $x = 0$.
- The x-**intercepts** are where the function's graph cuts the x-axis. These points are also called the **roots** (or **zeros**). To find them, set y equal to zero and solve for x.

Example

Sketch the graph of $y = -x^3 + x^2 + 2x$



Actual plot of $y = -x^3 + x^2 + 2x$



Rational Functions

Rational Functions

Rational Functions have the general form

$$f(x) = \frac{p(x)}{q(x)},$$

where $p(x)$ and $q(x)$ are polynomials.

- If degree of $p(x) <$ degree of $q(x)$,
  $f(x)$ is called a **strictly proper rational function**.
- If degree of $p(x) =$ degree of $q(x)$,
  $f(x)$ is called a **proper rational function**.
- If degree of $p(x) >$ degree of $q(x)$,
  $f(x)$ is called an **improper rational function**.

Rational Functions

An improper or proper rational function can always be expressed as a polynomial plus a strictly proper rational function, for example by algebraic division.

Example

$$\frac{4x^3 + 4x^2 + 4}{x^2 - 3} = 4x + 4 + \frac{12x + 16}{x^2 - 3}$$

For the previous example, we can work this out ourselves using **Long Division** to divide numerator by denominator:

Example

Use long division to show that

$$\frac{3x^4 + 2x^3 - 5x^2 + 6x - 7}{x^2 - 2x + 3} = 3x^2 + 8x + 2 - \frac{14x + 13}{x^2 - 2x + 3}$$

Try this yourself!

Towards Partial Fractions

Rational Functions have the general form $f(x) = \dfrac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials.

An (proper) rational function can often be written as a sum of simpler ones: **partial fractions.**

For example

$$\frac{8x - 12}{x^2 - 2x - 3}$$

can be written as

$$\frac{3}{x-3} + \frac{5}{x+1}$$

Check: (*next slide*)

Towards Partial Fractions

Exercises

Exercise 1.3.1

Sketch the graphs of

(i) $y = 5x^2 - 7$

(ii) $y = x^2 - 4x + 3$

(iii) $y = x^3 - 6x^2 - 11x - 6$